Project Guidelines or Student Project Description

Module- Automotive System Architecture: System-level Design

Target student: 1st year Engineering students in Automotive Software Engineering (**ASE1.1**)

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Learning Methodology: Project-Based Learning

(3h in-class + 3h autonomous work, over 2 weeks)

Semester 2

Project Title: Battery Energy Management (BEM) System

I. General Project goal

This project focuses on the system-level design of a Battery Energy Management (BEM) system within the context of an urban electric vehicle.

The BEM is a **safety-critical and performance-critical subsystem** that is strongly cross-coupled with multiple vehicle subsystems, including:

- Powertrain and inverter,
- Thermal management system,
- Vehicle Control Unit (VCU),
- Body electronics and HMI,
- Diagnostics and on-board communication infrastructure.

II. Customer Request: Battery Energy Management (BEM)

1. Project Context

The customer intends to develop a **new line of urban electric vehicles** equipped with a **78.1kWh lithium-ion battery pack**, configured as **96s46p (96 cells in series, 46 cells in parallel, similar to Tesla Model Y Long Range)**, using standard automotive-grade lithium-ion cells (3.7V nominal, ~5Ah capacity each).

2. Customer Needs

2.1 Operational Requirements

We need a battery system that:

1. **Works reliably in all climates**
   - Vehicle must start and operate in typical urban weather conditions.
2. **Provides accurate information to the driver**
   - Display remaining battery charge clearly (like a fuel gauge),
   - Show realistic driving range estimate,
   - Warn driver before battery becomes critically low,
   - Alert driver to any battery problems.
3. **Keeps the battery safe at all times**
   - Prevent fire or explosion under any circumstances,
   - Protect against electrical shock hazards,
   - Shut down safely if something goes wrong,
   - React quickly to dangerous situations,
4. **Maintains vehicle availability**
   - Vehicle should be available for use >99.9% of the time,
   - Minimize unexpected breakdowns,
   - Continue operating safely even when minor faults occur.
5. **Supports our charging infrastructure**
   - Compatible with standard charging stations
   - Protect battery during charging process

2.2 Integration with Vehicle Systems

The battery system must work with:

- **Vehicle controller** – receives power requests, coordinates vehicle functions,
- **Electric motor system** – delivers power for propulsion and receives regenerative braking energy,
- **Thermal management** – keeps battery at optimal temperature,
- **Dashboard** – displays battery status, warnings, and alerts to driver,
- **Charging system** – interfaces with external chargers (AC and DC fast charging),
- **Diagnostics** – allows service technicians to check battery health and troubleshoot problems.

2.3 Safety Expectations

The customer expects:

- **Fire and explosion prevention** – This is our top safety priority,
- **Electric shock protection** – High voltage (>350V) must be safely isolated,
- **Early warning of problems** – Driver should be alerted before dangerous situations develop,

- **Safe failure modes** – If something breaks, vehicle should stop safely (not catch fire or lose all power suddenly),
- **Professional service support** – Technicians can diagnose and repair issues using standard diagnostic tools.

2.4 User Experience Requirements

The customer wants:

1. **Predictable range** – "If it says 100 km remaining, I should be able to drive ~100 km",
2. **No surprises** – No unexpected shutdowns or sudden power loss,
3. **Clear communication** – Easy to understand warnings and status indicators,
4. **Long battery life** – Battery should last the lifetime of the vehicle (10+ years) with minimal degradation,
5. **Cold weather operation** – Vehicle should work in winter (even if range is reduced),

2.5 Regulatory Compliance

The battery system must comply with:

- Automotive safety standards (ISO 26262 for functional safety),
- Electrical safety regulations,
- On-board diagnostics requirements (OBD-II),
- Electromagnetic compatibility standards,
- Environmental regulations (operating conditions, recyclability).

3. Technical Context

Battery Pack Configuration:

- 96 cells in series × 46 cells in parallel = 4,416 total cells,
- Nominal voltage: ~355V,
- Total capacity: ~220 Ah,
- Total energy: ~78 kWh.

Vehicle Architecture: The vehicle uses distributed electronic control units communicating via CAN bus network. The battery system will be one of several major systems that must coordinate their operation.

III. System Engineering Approach

Students are expected to:

1. Analyze the customer needs in the context of the complete vehicle,
2. Translate customer needs into technical requirements,
   examples:
   - Convert "works in all climates" into specific temperature ranges.
   - Convert "accurate information" into measurement accuracy specifications.
   - Convert "react quickly" into response time requirements.
3. Design the system architecture :
   a. Divide the system into manageable subsystems (Consider functional and non-functional requirements)
   b. Define interfaces between sub-systems
   c. Allocate requirements to:
      - Hardware (sensors, circuits, components),
      - Software (algorithms, diagnostics, communication),
      - Mechanical aspects.

IV. Project Organization and Student Work Description

1. Proposed sub-projects:

The BEM project is organized into four technical sub-projects, each assigned to one student group:

- **Group 1 – Battery Monitoring & Data Acquisition.**
- **Group 2 – Battery State Estimation.**
- **Group 3 – Safety, Protection & Diagnostics.**
- **Group 4 – Communication & System Integration.**

Despite this division, the project follows a system engineering approach:

- All groups work on a shared vehicle-level architecture
- All requirements must be consistent and traceable at system level

2. WEEK 1 – Common System-Level Understanding (All Groups)

a) Objective of Week 1

The objective of Week 1 is to build a shared system-level understanding of the vehicle and the Battery Energy Management (BEM) system.

This week is a preliminary step before: Customer Needs Analysis and Requirement definition and traceability (Week 2)

No detailed requirements or design are expected in Week 1.

b) Deliverables – Week 1

Each group shall submit **one short document (4–6 pages max)** including the following parts:

1. Vehicle & System Context

- Type of vehicle and intended usage
- Battery pack overview (architecture, energy, voltage)
- Main operating conditions (driving, charging, environment)

Remark: understand the context, not define requirements.

2. Vehicle-Level Architecture

- One clear vehicle-level architecture diagram
- Position of the BEM inside the vehicle
- Main interactions between vehicle subsystems

Remark: No internal software or hardware design.

3. BEM Role and System Boundaries

- What the BEM is responsible for
- What is outside the BEM scope
- Vehicle-level impact of BEM decisions

Remark:Think in terms of responsibilities.

4. Main Interfaces and Interactions

- Interfaces between the BEM and other vehicle systems
- Information exchanged (no signal details)
- Safety or availability dependencies (qualitative)

Remark: This prepares interface-based requirements.

5. High-Level Functional View

- List of high-level BEM functions
- Short description of each function

Remark: These are not requirements yet.

3. WEEK 2 – Customer Needs & System Requirements (Group-Oriented)

a) Objective of Week 2

Week 2 focuses on formalizing customer needs and transforming them into system-level requirements, while starting to reflect the four technical sub-projects.

Building on the system understanding established in Week 1, students will:

- Capture and structure customer needs
- Derive clear and verifiable system requirements
- Start specializing requirements according to the four technical sub-projects
- Establish initial **requirement traceability**

b) Common Tasks for All Groups – Week 2

All groups shall work collaboratively to define a shared system baseline, including:

- Identification of **stakeholders** (driver, service, manufacturer, regulatory bodies)
- Formalization of **customer needs**
- Definition of **top-level BEM system requirements**
- Creation of a **common requirement traceability structure**

c) Group-Specific Focus – Week 2

After defining the common baseline, each group shall **refine and extend** the system requirements according to its assigned technical domain.

Group 1 – Battery Monitoring & Data Acquisition

Focus on system requirements related to:

- Cell voltage measurement
- Pack current measurement

- Temperature sensing
- Sampling rates and measurement accuracy
- Sensor redundancy and plausibility checks

Expected outputs (Week 2):

- Monitoring-related system requirements
- Traceability between customer needs and measurement constraints

Group 2 – Battery State Estimation

Focus on system requirements related to:

- State of Charge (SoC)
- State of Health (SoH)
- State of Power (SoP)
- Data dependencies and update rates
- Accuracy and robustness constraints

Expected outputs (Week 2):

- Estimation-related system requirements
- Defined interfaces with monitoring functions and vehicle control

Group 3 – Safety, Protection & Diagnostics

Focus on system requirements related to:

- Electrical and thermal protection
- Fault detection and classification
- Reaction times and safe responses
- Diagnostic coverage
- ISO 26262 safety constraints (ASIL C assumption)

Expected outputs (Week 2):

- Safety and protection system requirements
- Fault reactions and safe-state assumptions

Group 4 – Communication & System Integration

Focus on system requirements related to:

- CAN communication
- Diagnostic communication (OBD-II)
- BEM operating modes
- Coordination with VCU and other ECUs

Expected outputs (Week 2):

- Communication and integration system requirements
- Interface definitions and timing constraints

d) Deliverables – Week 2

Each group shall submit a **single structured document** containing:

- **Customer Needs Description**
    - Common structure and shared understanding
    - Clear link to stakeholder expectations
- **System-Level Requirements**
    - Derived from customer needs
    - Refined according to group technical focus
- **Requirement Traceability**
    - Customer needs → System requirements
    - Identification of group ownership for each requirement